

Glossary of Key Terms

Improving Deep Neural Networks

AI Foundations Series — Book 2

This glossary collects the main terms and concepts introduced in Book 2 — the practical toolkit for taking a network from “it runs” to “it works well.” Entries are arranged alphabetically and worded the way the book uses them: optimization methods (Momentum, RMSprop, Adam), regularization (L2, L1, dropout), normalization and initialization, the bias–variance lens, and hyperparameter tuning. Beginners can read it straight through; reviewers can jump to any entry.

A

Activation

The output of a layer's neurons after the nonlinearity (ReLU, sigmoid, and so on) is applied, written a . Contrast with the pre-activation z .

Adam (Adaptive Moment Estimation)

The default optimizer for most networks: it combines Momentum's gradient average with RMSprop's per-parameter squared-gradient scaling, both bias-corrected. Typical defaults $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1e-8$.

B

Batch Normalization

Normalizing the pre-activations of a hidden layer to a steady mean and variance during training, then rescaling with learnable parameters gamma and beta. Speeds up learning, eases initialization sensitivity, and mildly regularizes.

Beta (β)

The smoothing coefficient in an exponentially weighted average, between 0 and 1; a higher beta averages over more past values. Note: batch norm's beta is a different, learnable shift parameter that happens to share the letter.

Bias (high)

Underfitting: the model is too simple to capture the pattern, so even the training error is high. Diagnosed from a high train error.

Bias Correction

A division by $(1 - \beta^t)$ that fixes the cold-start undershoot of an exponentially weighted average during its first steps. Built into Adam.

Bias–Variance Tradeoff

The balance between underfitting (high bias) and overfitting (high variance). Reading train vs. dev error tells you which problem dominates and what to try next.

C

Coarse-to-Fine Search

A hyperparameter strategy: sample broadly first, then zoom in and sample more densely around the region that performed best.

Cost Function

A single number measuring how wrong the network currently is, averaged over examples. Lower is better; training minimizes it. This book uses cross-entropy cost for classification.

Covariate Shift

When the distribution of a model's inputs changes, forcing it to re-adapt even if the underlying rule is unchanged. Inside a deep network, batch norm reduces the covariate shift the later layers experience.

Cross-Entropy

The loss used for classification, measuring the distance between predicted probabilities and the true labels.

D

Data Augmentation

Expanding the training set with label-preserving transformations of existing examples (flips, crops, and so on) to reduce overfitting at little cost.

Decay (Learning-Rate)

Shrinking the learning rate over time — big steps early for fast progress, small steps late to settle into the minimum cleanly.

Dev Set

The development (validation) set: held-out data used to tune hyperparameters and judge progress, separate from both train and test sets.

Dropout

A regularizer that randomly silences a fraction of neurons each training step (keep-probability `keep_prob`), so the network cannot rely on any single unit. Turned off at test time; implemented with inverted dropout.

E

Early Stopping

Halting training when dev-set performance stops improving, used as a simple regularizer to prevent the network from over-learning the training data.

Epoch

One full pass through the entire training set. Contrast with an iteration, which is one update on one mini-batch.

Epsilon (ϵ)

A tiny constant added to a denominator (in RMSprop and Adam) to avoid dividing by zero and keep the update numerically stable.

Exploding Gradients

Gradients that grow uncontrollably as they propagate through many layers, destabilizing training. See Vanishing / Exploding Gradients.

Exponentially Weighted Average

A running average that weights recent values more than old ones: $v = \beta v + (1-\beta) \cdot \text{new}$. The building block under Momentum, RMSprop, and Adam.

G

Gamma (γ)

The learnable scale parameter in batch normalization; it sets the variance (spread) of a normalized hidden unit. Paired with beta, the learnable shift.

Gradient Checking

Numerically verifying analytic gradients with a two-sided finite difference, to catch silent backprop bugs before trusting the network. Turned off during normal training.

Gradient Tape

A framework feature (e.g. TensorFlow) that records the forward computation so gradients can be obtained automatically by replaying it backward — so you write only forward prop.

H

He Initialization

A weight-initialization scheme scaled for ReLU networks that keeps signal alive layer to layer, preventing vanishing or exploding gradients at the start of training.

Hyperparameter

A value you set before training that governs how learning happens (learning rate, batch size, number of layers, and so on). Contrast with a parameter, which the network learns.

I

Iteration

One parameter update on one mini-batch. Many iterations make up an epoch.

L

L1 Regularization

Adding a penalty proportional to the absolute size of the weights, which tends to push some weights to exactly zero (sparsity).

L2 Regularization

Adding a penalty proportional to the squared size of the weights to the cost, which shrinks weights (“weight decay”) and reduces variance.

Lambda (λ)

The regularization parameter controlling how strongly the L2 (or L1) penalty pulls weights toward zero. Larger lambda means stronger regularization.

Learning Rate (α)

The step size: how far each update moves the parameters against the gradient. The single most important hyperparameter to tune.

M

Mini-Batch

A small subset of the training data (often 64–256 examples) used for one gradient update, letting training take many steps per epoch. Sizes are typically powers of two.

Momentum

An optimizer that steps along an exponentially weighted average of the gradients (the velocity v), damping oscillations and accelerating along the consistent direction.

N

Normalization (Inputs)

Subtracting the mean and scaling the variance of the input features so they share a common scale, which rounds the cost surface and speeds learning. Dev and test data use the training statistics.

O

Optimizer

The rule that turns gradients into a parameter update — plain gradient descent, Momentum, RMSprop, or Adam.

Overfitting

When a model learns the training set too well, including its noise, and generalizes poorly. The same as high variance.

P

Parameter

A value the network learns from data during training — the weights W , biases, and batch norm's gamma and beta. Contrast with a hyperparameter.

Pre-activation (z)

The weighted sum $W a + b$ computed by a layer before the nonlinearity is applied. Batch norm normalizes z , not the activation a .

R

Random Search

Sampling hyperparameter combinations at random rather than on a fixed grid, which explores the important dimensions more efficiently than grid search.

RMSprop

An optimizer that divides each parameter's step by a running root-mean-square of its recent gradients, damping noisy directions and speeding steady ones.

S

Softmax

The output activation for multi-class classification: it turns a vector of scores into positive probabilities that sum to one. Generalizes logistic regression to C classes.

Symmetry Breaking

The reason weights must be initialized randomly rather than to zero: identical starting weights would keep all neurons computing the same thing and never diverge.

T

TensorFlow

The deep-learning framework introduced late in the book; its gradient tape lets you define only the forward computation and obtain gradients automatically.

Test Set

Held-out data used once, at the end, to estimate real-world performance. Never tuned against.

Train Set

The data the network learns from by updating its parameters; train error is compared against dev error to diagnose bias and variance.

V

Variance (high)

Overfitting: the model fits the training data far better than the dev data, shown by a low train error but a high dev error.

Vanishing / Exploding Gradients

Gradients that shrink toward zero or blow up as they propagate through many layers, stalling or destabilizing training. Mitigated by careful initialization and batch norm.

Velocity (v)

The exponentially weighted average of the gradients that Momentum (and Adam) steps along.

W

Weight Decay

The shrinking effect L2 regularization has on the weights at each update; equivalently, the penalty that pulls weights toward zero.

X

Xavier Initialization

A weight-initialization scheme scaled for tanh (and similar) networks, keeping the variance of signals stable across layers. The tanh-network counterpart to He initialization.